

Problem

Construct the Bode plots for

$$H(s) = \frac{10}{s(s^2 + 80s + 400)}$$

Step-by-step solution

Step 1 of 3

Given transfer function is,

$$H(s) = \frac{10}{s(s^2 + 80s + 400)}$$

Put $s = j\omega$

$$H(\omega) = \frac{10}{j\omega((j\omega)^2 + 80j\omega + 400)}$$

$$H(\omega) = \frac{10}{400(j\omega)\left(1 + \frac{80}{400}j\omega - \frac{\omega^2}{400}\right)}$$

$$H(\omega) = \frac{1}{40(j\omega)\left(1 + j\omega 0.2 - \left(\frac{\omega}{20}\right)^2\right)}$$

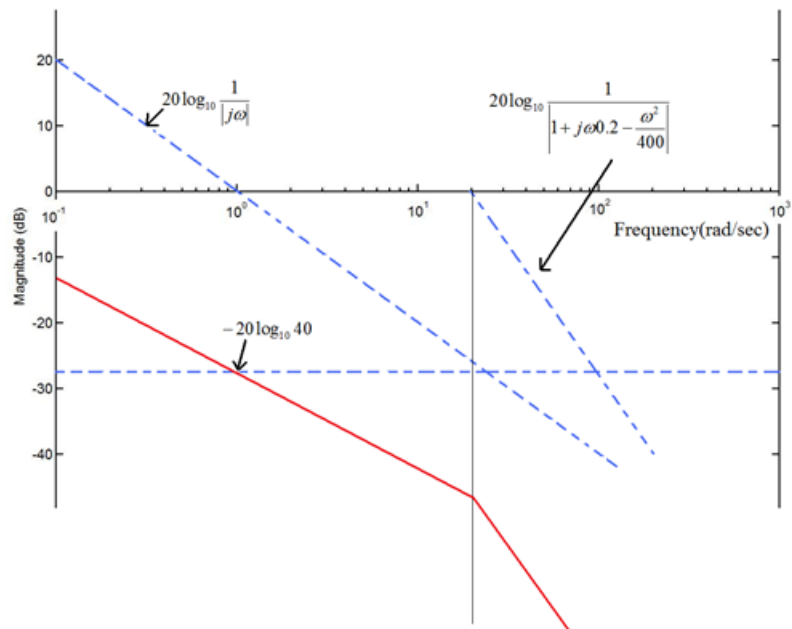
Hence, the magnitude and phase are

$$H(\omega) = -20 \log_{10} 40 - 20 \log_{10} (\omega) - 20 \log_{10} \sqrt{\left[1 - \left(\frac{\omega}{20}\right)^2\right]^2 + (0.2\omega)^2} \quad \dots\dots(1)$$

$$\phi = -90^\circ - \tan^{-1} \left(\frac{0.2\omega}{\left|1 - \frac{\omega^2}{400}\right|} \right) \quad \dots\dots(2)$$

The magnitude and phase poles are

Step 2 of 3



Step 3 of 3

